



RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

B.Sc. (General) Degree

First Year Semester II Examination – March / April 2014

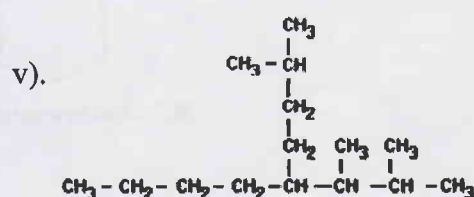
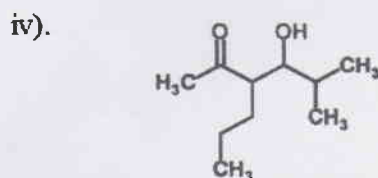
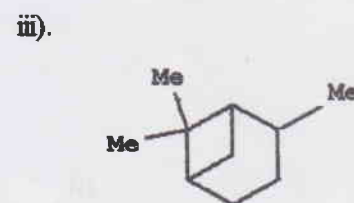
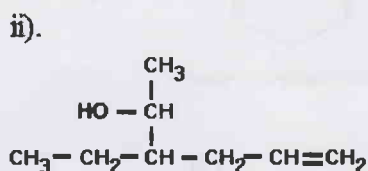
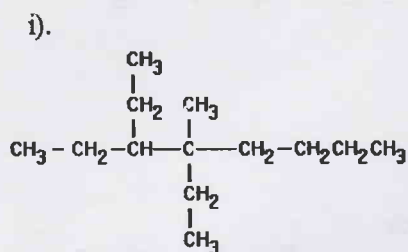
CHE 1203 – ORGANIC CHEMISTRY I

Answer any four questions.

Time: 02 hours

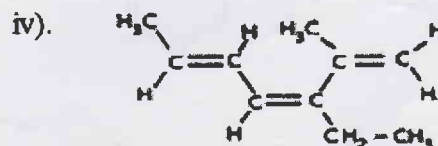
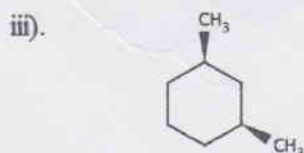
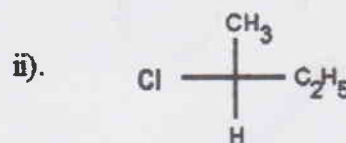
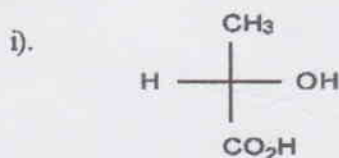
01.

(a). Give IUPAC names of the following compounds



(15 marks)

(b). Give IUPAC names of the following compounds with R/S, *cis/trans* or E/Z configurations



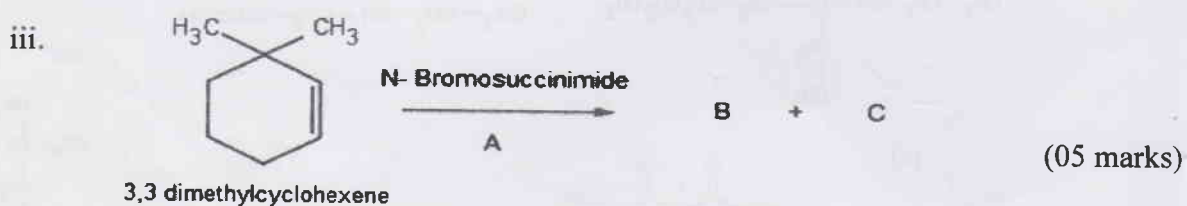
(12 marks)

(c). Draw the structure of the compound (Z)-3-chloro-4-methylhept-3-ene.

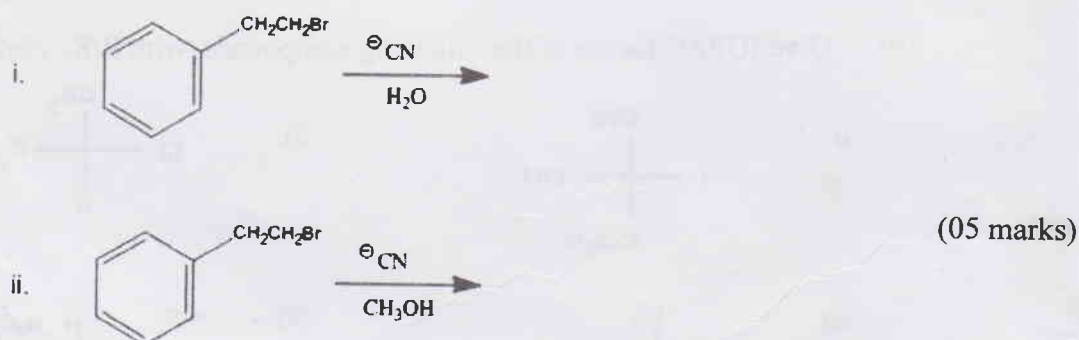
(03 marks)

02.

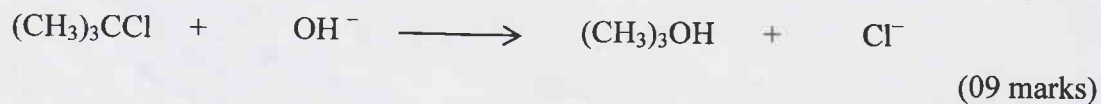
(a). Identify the reaction conditions, the products and complete the following reactions



(b). Complete the following equations and compare the rates of reaction with each other



(c). The given below is the nucleophilic substitution reaction of tert-butyl chloride and hydroxide ion to form tert-butyl alcohol and chloride ion. Write all the steps and the mechanism of this reaction and draw the relevant energy diagram.

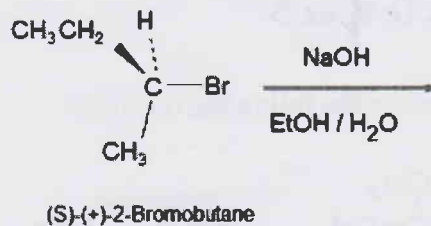


03.

- (a). Write a short note on the factors influencing the rate of S_N2 reaction

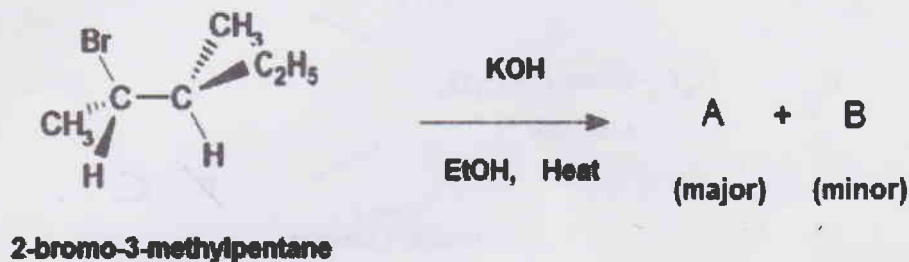
(05 marks)

- (b). Identify the intermediate & the final product, and discuss the stereoisomerism of this S_N2 reaction.



(06 marks)

- (c). Describe the products A and B of the following elimination reaction according to the Zaitsev's Rule.

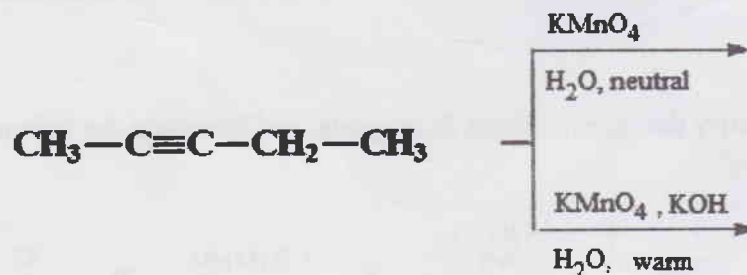


(06 marks)

- d). If the major product (A) of the above reaction in section (c) is having two isomers, draw and name the isomers produced by the above reaction when the reaction is E_2 .

(04 marks)

- (e). Identify the products and complete following reactions



(04 marks)

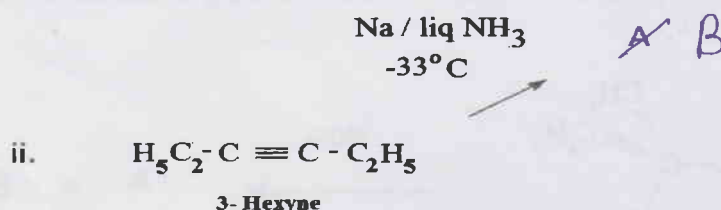
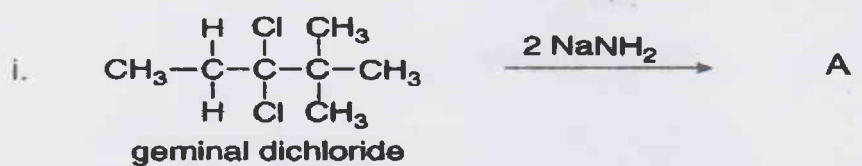
04.

- (a). Compare the carbon – carbon bond length of ethyne with the carbon – carbon bond lengths of respective ~~alkanes~~ and ~~alkanes~~.

alkenes alkynes

(04 marks)

- (b). Identify A, B and C and complete the following reactions



$\xrightarrow[\text{-33}^\circ\text{C}]{\text{Na / liq NH}_3}$

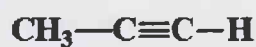
A B

$\xrightarrow[\text{Lindlar's catalyst}]{\text{H}_2/}$

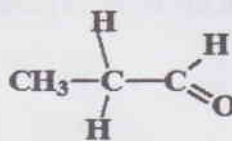
B C

(06 marks)

- (c). Outline all steps in the synthesis of 1-propanal from propyne using Sia_2BH (Disiamyl borane) in the initial step.



Propyne



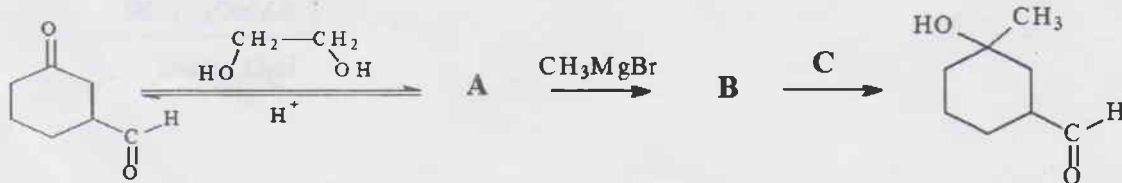
Propanal

(06 marks)

- (d). Describe the synthesis of benzaldehyde using Gatterman-Koch reaction.

(05 marks)

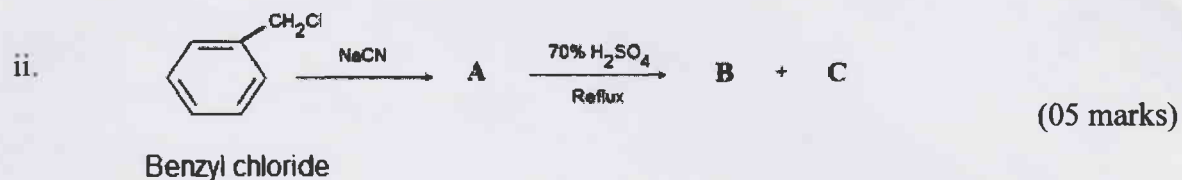
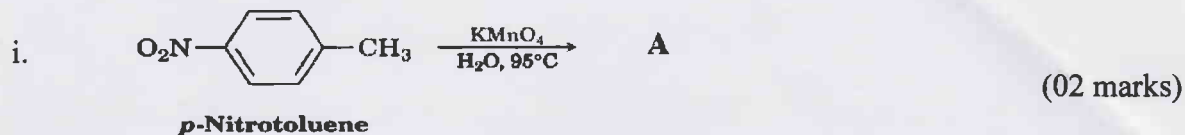
- (e). Identify the intermediates & reagents, and complete the following reaction



(04 marks)

05.

(a). Complete the following reactions



(b). “-NO₂ deactivates the benzene ring towards electrophilic aromatic substitution reactions and directs substitution to the meta- position”, explain the above sentence.

(06 marks)

(c). Discuss the synthesis of 1,4 -dibromo-2-nitrobenzene from benzene

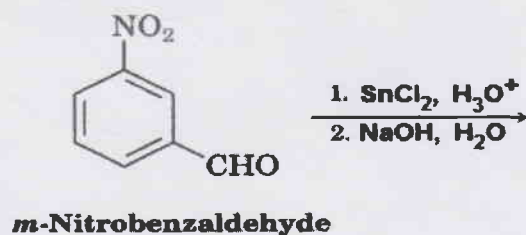
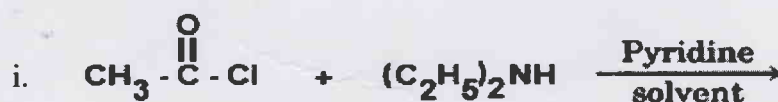


(05 marks)

(d). Reaction of NH₃ with CH₃OH in the presence of alumina catalyst produces a mixture of methylated amines. Outline the complete equation of the reaction.

(03 marks)

(e). Complete the following reactions



(04 marks)

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